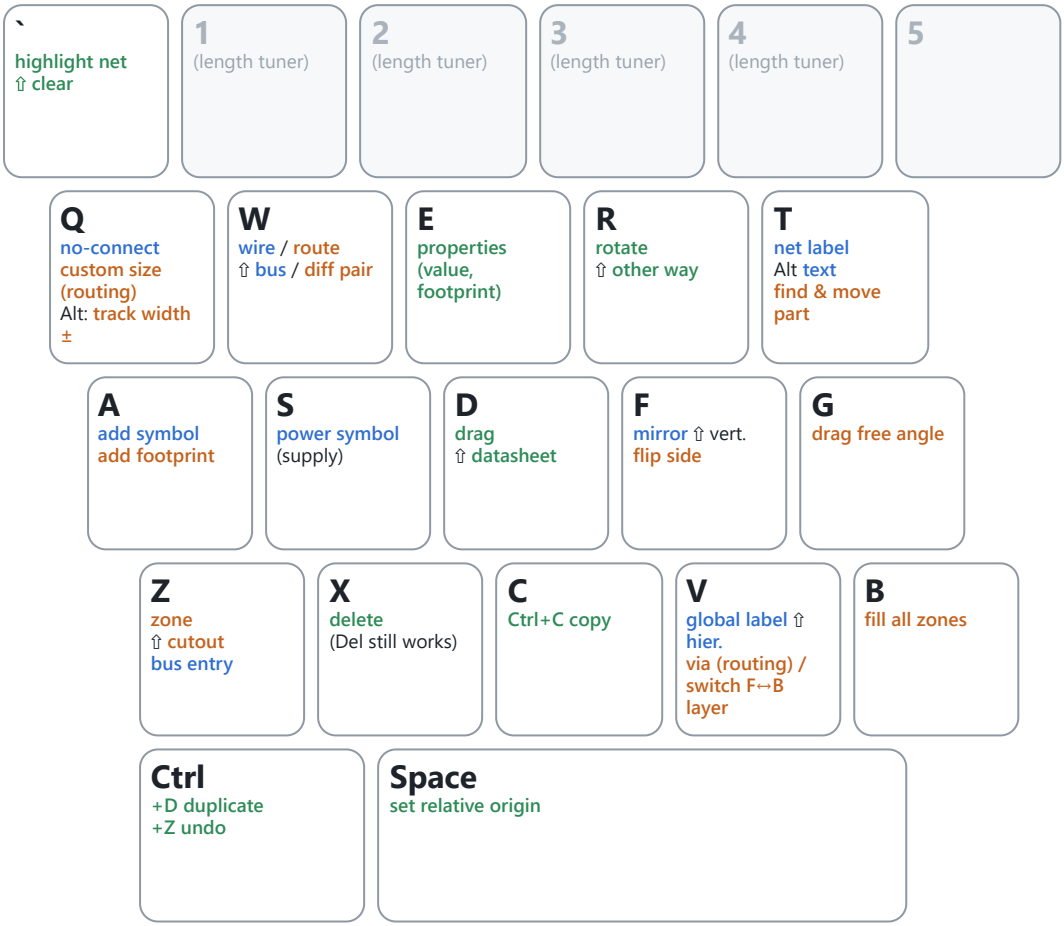
KiCad one-hand layout

Left hand on the keyboard, right hand on the mouse. The same key does the same *idea* in the schematic and the PCB editor.

Schematic PCB Both



The idea behind each key

Key	Idea	Schematic	PCB
W	connect	wire (⇧ bus)	route track (⇧ differential pair)
V	jump	global label (⇧ hierarchical): jumps between sheets	via: jumps between layers
T	name	net label (Alt+T free text)	find & move footprint
A S	add	symbol / power symbol	footprint
D M	move	drag keeps wires / move breaks them	drag keeps tracks / move
F	flip	mirror (⇧ vertical)	flip to the other side
X	remove	delete in both (Del still works)	
Z B	copper	—	zone, ⇧ cutout · B fills all zones

Install / undo

Install: Preferences › Preferences › Hotkeys › *Import Hotkeys…* › `onehand.hotkeys` (or copy it over `%APPDATA%\kicad\10.0\user.hotkeys` and restart KiCad). **Undo:** same dialog › *Reset to defaults*. It changes 20 of

890 bindings; everything else stays KiCad default.

Freed / moved: sheet (toolbar only), Edit Value & Edit Footprint (use E), grid ±stays N / ⇧N, track width ± is Alt+Q / Alt+⇧Q.

MyPlugin workflow

One toolbar button → hub with twelve tools (+ a Refresh Copper Paths button). Run them in this order.

#	Tool	What it does	When
1	Fab Rules	Board Setup constraints + Default netclass + DRC rules = Lion Circuits limits (MII 6/6 mil, 0.35 mm drill, 0.66 mm via…). Can set it as the default for every new project.	first, once per board
2	Net Setup	Volts + amps per net → IPC-2221 width, clearance, via size and count. Creates netclasses, fills the width/via dropdowns, writes DRC rules so thin power traces become DRC errors.	after schematic sync
3	Power Pours / Copper Path	Pours around regulator pad groups + plane vias. Copper Path: route the net (W), pick a width and 45°/round corners - wide copper that follows the traces, keeps clearance and narrows into pin rows; after dragging press *Refresh Copper Paths*.	after placement
4	Via Stitching	ViaStitcher (bundled): select the GND zone, fill it with a stitching-via grid.	after pours
5	Autoroute H/V	Freerouting: horizontal top, vertical bottom. Uses your netclasses; never shrinks a trace to make it fit.	signals
6	Power Trace	Select a pad/trace → V/A → netclass + widen existing traces → press `W` and drag.	power nets by hand
7	Silk Beautify	Reference placement, net names on connector pins, values to Fab; lists what it couldn't place.	last, before DRC
8	Docs (PcbDraw)	Front/back drawings + renders into 04_Docs/.	at the end
—	FreeCAD (StepUp)	Opens the board in FreeCAD with StepUp loaded; FreeCAD reloads when KiCad saves; Push buttons send changes back and KiCad reloads - no restarts.	enclosure work
—	3D Models	Matches each footprint without a model to a KiCad library footprint by pad geometry and attaches its 3D model.	after an Eagle import
—	Eagle → KiCad Symbols	Swaps imported Eagle symbols for KiCad ones; keeps Eagle pin numbers; refuses to save if the netlist changes.	after an Eagle import

Design blocks (KiCad 10)

A design block is a reusable circuit: a schematic snippet, and in KiCad 10 optionally its PCB layout too. Think "buck converter block", "USB-C input block", "ESP32 core block".

Make one

1. **Create a library once:** View › Panels › **Design Blocks**, then in the panel's menu make a new design block library (e.g. `MyBlocks`).
2. **Schematic part:** select the circuit (parts, wires, labels) › right-click › **Save Selection as Design Block**, pick `MyBlocks`, give it a name, description and keywords.

Use one

1. **Schematic:** drag the block from the Design Blocks panel. Options: repeated placement, place as group or as a new sheet, keep or re-annotate references.
2. **PCB:** Update PCB from Schematic (F8). Footprints that came from the block can take the saved layout (option *Apply design block*

3. **PCB part:** in the PCB editor place and route that same circuit, select it › right-click the **existing** block in the Design Blocks panel › **Update Design Block from Selection**. It must be the same block as the schematic, not just the same library.

layouts to new groups; keep *Group footprints* on).

3. Library location: `Preferences › Manage Design Block Libraries`.

> **Known issues (KiCad 10.0.x, community reports):** blocks with nested sheets are not supported; running *Update Design Block from Selection* again on a block that already has a linked group nests groups inside groups (check before repeating); very large blocks (100+ parts) applied many times have crashed the PCB editor. Save before using them.

Good first blocks for your roadmap: LM2596 5 V buck (from `01_Beginner/03`), AMS1117 3.3 V LDO, USB-C 5 V input with CC resistors, ESP32 auto-reset (UMH3N), PC817 relay driver.

Eagle ⇄ KiCad

Eagle → KiCad (works well)

1. **Board:** KiCad › File › Import › Non-KiCad Board › `.brd` (or `kicad-cli pcb import --format eagle`). Copper, outline, zones and nets come across.
2. **Schematic:** KiCad › File › Import › Non-KiCad Schematic › `.sch`. Eagle libraries get embedded as an `eagle_import` library.
3. **MyPlugin › 3D Models:** Eagle footprints have no 3D. This attaches the matching KiCad model by pad geometry (tested: 260 of 261 parts on a real board).
4. **MyPlugin › Eagle → KiCad Symbols:** replaces Eagle symbols with KiCad library symbols, keeps pin numbers so the PCB still matches, and verifies the netlist is identical before saving.
5. **MyPlugin › Fab Rules** then **Net Setup**: Eagle's DRC does not carry over cleanly; set rules again in KiCad.

KiCad → Eagle (no proper automatic path)

There is **no native KiCad → Eagle or KiCad → Fusion Electronics importer**. Autodesk forum threads have asked for one for years without it appearing. A paid service (Elgris) offers conversion; user reports on its quality are mixed.

You need…	Do this
Boards made	Send **Gerbers + drill** (or ODB++ / IPC-2581) from KiCad. Fabs never need Eagle files.
Someone using Eagle/Fusion to *view* it	Export PDF schematics, 3D STEP, Gerbers. Fusion opens STEP; Gerber viewers are everywhere.
To *edit* it in Eagle/Fusion	Re-enter it: import the outline as DXF (KiCad › File › Export › DXF), redraw the schematic with matching Eagle libraries, place parts from KiCad's position file, then route (a DSN/SES round-trip through Freerouting can carry traces, but only if part names and footprints match exactly).

> **Recommendation:** keep KiCad as the master copy. Autodesk has announced the retirement of standalone EAGLE, and every fab accepts Gerbers. Converting *into* KiCad is solid; converting *out of* it is a manual job.

Sources

Lion Circuits capabilities: lioncircuits.com/pcb-manufacturing-capabilities · KiCad add-on packaging: dev-docs.kicad.org/en/addons · Design blocks (community): forum.kicad.info threads "PCB design block from schematic", "KiCad 10.0.4 Design Block Problems", "layout editor crashes when clicking on a newly placed design block"; gitlab.com/kicad/code/kicad/-/issues/25368 · KiCad → Eagle: forums.autodesk.com "KiCad to EaglePCB" and "Importing KiCad projects" threads · Trace width: IPC-2221 ($I = k \cdot \Delta T^{0.44} \cdot A^{0.725}$).